17.5: Mutations of one or a few nucleotides can affect protein structure and function

Now that you have explored the process of gene expression, you can consider the effects of changes to the genetic information of a cell. Because these changes, called **mutations**, are the ultimate source of new genes, they are responsible for the huge diversity of genes found among organisms. Earlier, we considered chromosomal rearrangements that affect long segments of DNA (see Figure 15.14); these are considered large-scale mutations. Here we examine small-scale mutations of one or a few nucleotide pairs, including **point mutations**, changes in a single nucleotide pair of a gene.

If a point mutation occurs in a gamete or in a cell that gives rise to gametes, it may be transmitted to offspring and to future generations. If the mutation has an adverse effect on the phenotype of a person, the mutant condition is referred to as a genetic disorder or hereditary disease. For example, we can trace the genetic basis of sickle-cell disease to the mutation of a single nucleotide pair in the gene that encodes the β-globin polypeptide of hemoglobin. The change of a single nucleotide in the DNA's template strand leads to an altered mRNA and the production of an abnormal protein **(**Figure 17.26**; also see Figure 5.19). In individuals who are homozygous for the mutant allele, the sickling of red blood cells caused by the altered hemoglobin produces the multiple symptoms associated with sickle-cell disease (see Concept 14.4 and Figure 23.18). Another disorder caused by a point mutation is a heart condition called familial cardiomyopathy, which is responsible for some of the tragic incidents of sudden death in young athletes. Point mutations in several genes encoding muscle proteins have been identified, any of which can lead to this disorder.

Figure 17.26

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    The molecular basis of sickle-cell disease: a point
mutation.
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The molecular basis of sickle-cell disease.

The allele that causes sickle-cell disease differs from the wild-type (normal) allele by a single DNA nucleotide pair. The micrographs are SEMs of a normal red blood cell (on the left) and a sickled red blood cell (right) from individuals homozygous for wild-type and mutant alleles, respectively.

Wild type D N A with partial sequence shown. Template strand reads 3 prime C T C 5 prime; nontemplate strand reads 5 prime G A C 3 prime. Normal m R N A transcribed from template strand reads 5 prime G A G

3 prime. Translation of the normal m R N A results in a normal hemoglobin with glutamic acid (Glu) translated for the m R N A codon shown. Result is normal round red blood cells.

Sickle-cell beta-globin. Transcription of mutant beta-globin D N A (template has an A where there should be a T) results in mutant m R N A and sickle-cell hemoglobin. Result is sickle-shaped red blood cells. In the D N A, the mutant (sickle-cell) template strand (top) has an A where the wild type template has a T. Mutant D N A template strand reads: C A G. After transcription, the mutant m R N A has a U instead of an A in one codon. The mutant codon reads G U G. After translation, the mutant beta-globin has a valine (corresponds to the m R N A codon G U G) instead of a glutamic acid (Glu). The result is sickle-shaped red blood cells.

HHMI BioInteractive Animation

Sickle-Cell Disease

image

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Types of Small-Scale Mutations

Many mutations occur in regions outside of protein-coding genes, and any potential effect they have on the phenotype of the organism may be subtle and hard to detect. For this reason, here we'll concentrate on mutations within protein-coding genes. Small-scale mutations within a gene can be divided into two general categories: (1) single nucleotide-pair substitutions and (2) nucleotide-pair insertions or deletions. Insertions and deletions can involve one or more nucleotide pairs.

Substitutions

A **nucleotide-pair substitution** is the replacement of one nucleotide and its partner with another pair of nucleotides **(Figure 17.27a)**. Some substitutions have no effect on the encoded protein, owing to the redundancy of the genetic code. For example, if 3′-CCG-5′ on the template strand mutated to 3′-CCA-5′, the mRNA codon that used to be GGC would become GGU, but a glycine would still be inserted at the proper location in the protein (see Figure 17.6). In other words, a change in a nucleotide pair may transform one codon into another that is translated into the same amino acid. Such a change is an example of a **silent mutation**, which has no observable effect on the phenotype. (Silent mutations can occur outside genes as

well.) Interestingly, there is evidence that some silent mutations may indirectly affect where or at what level the gene gets expressed, even though the actual protein is the same.

Figure 17.27

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Types of small-scale mutations that affect mRNA sequence.
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Wild type. Transcription of D N A template strand leads to normal m R N A and a normal protein (here: M e t, Lys, P h e, G l y).

All but one of the types shown here also affect the amino acid sequence of the encoded polypeptide.

Transcriptions are as follows.

Wild type D N A Template strand: 3 prime T A C T T C A A A C C G A T T 5 prime. Nontemplate strand: 5 prime-A T G A A G T T T G G C T A A -3 prime. m R N A: 5 prime-A U G A A G U U U G G C U A A-3 prime. Final codon (U A A) is a stop codon. Protein: (Amino end) Met-Lys-Phe-G l y (Carboxyl end).

Nucleotide-pair substitution: Silent. Underlined components differ from the wild type. An A is substituted for the final G in the template D N A, so that the template strand now reads: 3 prime T A C T T C A A A C C underlined A, A T T 5 prime. Resulting m R N A reads: 5 prime A U G A A G U U U G G underline U U A A-3 prime. UAA is a stop codon. Protein: M e t, L y s, P h e, G l y. This mutation is silent. There is no effect on the amino acid sequence (same as wild type protein).

Nucleotide pair substitution: Missense. Underlined components differ from the wild type. T is substituted for C in template D N A strand, which now reads: 3 prime T A C T T C A A A underlined T C G A T T 5 prime. Result is A instead of G in one m R N A codon: 5 prime A U G A A G U U U underlined A G C U A A 3 prime. U A A is a stop codon. Protein: M e t, Ly s, P h e, underlined S e r. Result is a range of effects depending on the location within the protein and the identity of the new amino acid.

Nucleotide pair substitution: Nonsense. Underlined components differ from the wild type. A is substituted for T in template D N A strand, which now reads: 3 prime T A C underlined A T C A A A C C G A T T 5 prime. Result is U instead of A in m R N A: 5 prime A U G underlined U A G U U U G G U U A A 3 prime. U A G is a stop codon. Protein: Met Stop. Nonsense: Effect depends on how close the mutation is to the wild-type stop codon.

Nucleotide pair insertion: Frameshift. Underlined components differ from the wild type. Extra A is inserted between first two codons of D N A template, which now reads: 3 prime T A C underlined A T T C A A A C C G A T T 5 prime. Result is an extra U in the m R N A: 5 prime A U G underlined U A A G U U U G G C U A A 3

prime. The extra U creates a stop codon, U A A in the second reading frame. Protein: Met. Frameshift causes immediate nonsense after one nucleotide pair insertion.

Nucleotide pair deletion: Frameshift. Underlined components differ from the wild type. An A is missing from the D N A template, which now reads: 3 prime T A C T T C A A C C G A T T 5 prime. The missing A was ninth in the sequence in the wild type template. Result is a missing U in the m R N A: 5 prime A U G A A G U U G G C U A A 3 prime. The missing U would have been ninth in this sequence. Protein: M e t, L y s, underlined L e u, underlined A l a. Frameshift causes extensive missense (1 nucleotide pair deletion).

3 Nucleotide pair deletion. Underlined components differ from the wild type. 3 nucleotides are deleted from template D N A, which now reads: 3 prime T A C A A A C C G A T T 5 prime. The bases T T C are missing and should follow T A C. Result is a missing codon in the m R N A: 5 prime A U G U U U G G C U A A 3 prime. Codon A A G is missing and should follow A U G. UAA is a stop codon. Protein: M e t, P h e, G l y.

3 nucleotide-pair deletion: No frameshift, but one amino acid is missing. A 3-nucleotide-pair insertion (not shown) would lead to an extra amino acid.

Figure 17.27 Walkthrough

Types of small-scale mutations that affect mRNA sequence

Watch an explanation of this figure in the Figure 17.27 Walkthrough: Types of small-scale mutations that affect mRNA sequence.

image

Play

image

Play

Animation

Mutation Types

image

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Substitutions that change one amino acid to another one are called **missense mutations**. Such a mutation may have little effect on the protein: The new amino acid may have properties similar to those of the amino

acid it replaces, or it may be in a region of the protein where the exact sequence of amino acids is not essential to the protein's function.

However, the nucleotide-pair substitutions of greatest interest are those that cause a major change in a protein. The alteration of a single amino acid in a crucial area of a protein—such as in the part of the β-globin subunit of hemoglobin shown in Figure 17.26 or in the active site of an enzyme as shown in Figure 8.19—can significantly alter protein activity. Occasionally, such a mutation leads to an improved protein or one with novel capabilities, but much more often such mutations are neutral or detrimental, leading to a useless or less active protein that impairs cellular function. A second example of a missense mutation is a mutation in the tyrosinase gene that causes the albino phenotype in the Asinara donkey. Recently, Italian researchers sequenced the tyrosinase gene in donkeys from this wild population and showed that a recessive mutation is responsible for the lack of pigment. In this gene, a C is changed to a G, resulting in a histidine being added to the protein instead of an aspartic acid. The changed amino acid is in a site in tyrosinase that normally binds a copper atom, and it makes the enzyme incapable of binding copper, rendering it nonfunctional. As in the case of sickle-cell disease, just one change in a nucleotide base results in a dramatic phenotypic change.

Substitution mutations are usually missense mutations; that is, the altered codon still codes for an amino acid and thus makes sense, although not necessarily the *right* sense. But a point mutation can also change a codon for an amino acid into a stop codon. This is called a **nonsense mutation**, and it causes translation to be terminated prematurely; the resulting polypeptide will be shorter than the polypeptide encoded by the normal gene. Most nonsense mutations lead to nonfunctional proteins.

In the **Problem-Solving Exercise**, you'll work with a few common single nucleotide-pair substitution mutations in the gene encoding insulin, some or all of which may lead to diabetes. You will classify these mutations into one of the types we just described and characterize the change in the amino acid sequence.

Problem-Solving Exercise

AP

Are insulin mutations the cause of three infants' neonatal diabetes?

A healthcare professional holds a newborn close to her body.

Insulin is a hormone that acts as a key regulator of blood glucose level, and insulin deficiency can lead to diabetes. In some cases of neonatal (newborn) diabetes, the gene coding for the insulin protein has a nucleotide-pair substitution mutation that alters the protein structure enough to cause it to malfunction.

Doctors can use DNA sequence information to diagnose diabetes and other diseases. For example, the insulin gene sequence of a patient with neonatal diabetes can be analyzed to determine if it has a mutation and, if so, its effect.

In this exercise, you will determine the effect of mutations within insulin gene sequences in infant diabetes patients.

Your Approach Suppose you are a medical geneticist presented with three infant patients, all of whom have a nucleotide-pair substitution in their insulin gene. It is your job to analyze each mutation to figure out its effect on the amino acid sequence of the insulin protein. To identify the mutation in each patient, you will compare his or her individual insulin complementary DNA (cDNA) sequence to that of the wild-type cDNA. (cDNA is a double-stranded DNA molecule that is based on the mRNA sequence and thus contains only the portion of a gene that is translated—introns are not included. cDNA sequences are commonly used to compare the coding regions of genes.) Identifying the codons that have been changed will tell you which, if any, amino acids are altered in the patient's insulin protein.

Your Data You will analyze the cDNA codons for amino acids 35–54 (of the 110 amino acids) of each patient's insulin protein, so the start codon (AUG) is not present. The sequences of the wild-type cDNA and the patients' cDNA are shown below, arranged in codons.

A table provides the c D N A sequences of the wild type and of patients 1, 2, and 3.

The table is as follows.

Gene (codes for enzymes)	Wild type	Class Roman numeral 1 mutants (mutation in gene A)	Class Roman numeral 2 mutants (mutation in gene B)	Class Roman numeral 3 mutants (mutation in gene C)
Gene A	Enzyme A (Precursor goes to Ornithine)	Enzyme A crossed out in the step Precursor goes to Ornithine.	Enzyme A (Precursor goes to Ornithine)	Enzyme A (Precursor goes to Ornithine)
Gene B	Enzyme B (Ornithine goes to Citrulline)	Enzyme B (Ornithine goes to Citrulline)	Enzyme B crossed out in the step Ornithine goes to Citrulline)	Enzyme B (Ornithine goes to Citrulline)

Gene C	Enzyme C (Citrulline goes to Arginine)	Enzyme C (Citrulline goes to Arginine)	Enzyme C (Citrulline goes to Arginine)	Enzyme C crossed out in the step Citrulline goes to Arginine.

Data from N. Nishi and K. Nanjo, Insulin gene mutations and diabetes, *Journal of Diabetes Investigation* 2:92–100 (2011).

Your Analysis

- Compare each patient's cDNA sequence to the wild-type cDNA sequence. Circle the codons where a nucleotide-pair substitution mutation has occurred.
- Use a codon table (see Figure 17.6) to identify the amino acid encoded by the codon with the mutation in each patient's insulin sequence, and compare it to the amino acid encoded by the codon in the corresponding wild-type sequence. As is standard practice with DNA sequences, the *coding* (nontemplate) strand of the cDNA has been provided, so to convert it to mRNA for use with the codon table, you only need to change T to U. Classify each patient's nucleotide-pair substitution mutation: Is it a silent, missense, or nonsense mutation? Explain, for each answer.
- Compare the structure of the amino acid you identified in each patient's insulin sequence to that of the corresponding amino acid in the wild-type sequence (see Figure 5.14). Given that each patient has neonatal diabetes, discuss how the change of amino acid in each might have affected the insulin protein and thus resulted in the disease. (Consider the chemical nature of the side chains.)

Instructors: A version of this Problem-Solving Exercise can be assigned in **Mastering Biology**. Or a more extensive investigation called "Solve It: Which Insulin Mutations May Result in Disease?" can be assigned.

Insertions and Deletions

Insertions and **deletions** are additions or losses of nucleotide pairs in a gene **(Figure 17.27b)**. These mutations have a disastrous effect on the resulting protein more often than substitutions do. Insertion or deletion of nucleotides may alter the reading frame of the genetic message, the triplet grouping of nucleotides on the mRNA that is read during translation. Such a mutation, called a **frameshift mutation**, occurs whenever the number of nucleotides inserted or deleted is not a multiple of three. All nucleotides downstream of the deletion or insertion will be improperly grouped into codons; the result will be extensive missense mutations, usually ending sooner or later in a nonsense mutation that leads to premature termination. Unless the frameshift is very near the end of the gene, the protein is almost certain to be

nonfunctional. Insertions and deletions also occur outside of coding regions; these are not called frameshift mutations, but can have effects on the phenotype—for instance, they can affect how a gene is expressed.

New Mutations and Mutagens

Mutations can arise in a number of ways. Errors during DNA replication or recombination can lead to nucleotide-pair substitutions, insertions, or deletions, as well as to mutations affecting longer stretches of DNA. If an incorrect nucleotide is added to a growing chain during replication, for example, the base on that nucleotide will then be mismatched with the nucleotide base on the other strand. In many cases, the error will be corrected by DNA proofreading and repair systems (see Concept 16.2). Otherwise, the incorrect base will be used as a template in the next round of replication, resulting in a mutation. Such mutations are called *spontaneous mutations*. It is difficult to calculate the rate at which such mutations occur. Rough estimates have been made of the rate of mutation during DNA replication for both *E. coli* and eukaryotes, and the numbers are similar: About one nucleotide in every 1010 is altered, and the change is passed on to the next generation of cells.

A number of physical and chemical agents, called **mutagens**, interact with DNA in ways that cause mutations. In the 1920s, Hermann Muller discovered that X-rays caused genetic changes in fruit flies, and he used X-rays to make *Drosophila* mutants for his genetic studies (just as Beadle and Tatum did with their *Neurospora* cells). But Muller also recognized an alarming implication of his discovery: X-rays and other forms of high-energy radiation pose hazards to the genetic material of people as well as laboratory organisms. Mutagenic radiation, a physical mutagen, includes ultraviolet (UV) light, which can cause disruptive thymine dimers in DNA (see Figure 16.20).

Chemical mutagens fall into several categories. Nucleotide analogs are chemicals similar to normal DNA nucleotides but that pair incorrectly during DNA replication. Other chemical mutagens interfere with correct DNA replication by inserting themselves into the DNA and distorting the double helix. Still other mutagens cause chemical changes in bases that change their pairing properties.

Researchers have developed a variety of methods to test the mutagenic activity of chemicals. A major application of these tests is the preliminary screening of chemicals to identify those that may cause cancer. This approach makes sense because most carcinogens (cancer-causing chemicals) are mutagenic, and conversely, most mutagens are carcinogenic.

Using CRISPR to Edit Genes and Correct Disease-Causing Mutations

Ever since biologists have understood how disease-causing proteins were the result of mutations in genes, they have sought techniques for **gene editing**—altering genes in a specific, predictable way. Their aim has

been to use such a technique to change specific genes in living cells, in part to try to correct genes that cause disease. Over the past 15 years, biologists have developed a powerful new technique for gene editing, called the **CRISPR-Cas9 system**, that is transforming the field of *genetic engineering*, the direct manipulation of genes for practical purposes. Cas9 is a bacterial protein that helps defend bacteria against the viruses that infect them (bacteriophages). In bacterial cells, Cas9 acts together with "guide RNA" made from the CRISPR region of the bacterial genome. (Figure 19.7 explains how this defense system works in bacteria.)

HHMI BioInteractive Video

Jennifer Doudna: CRISPR Provides a Tool for Changing Genes

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Interview

Interview with Francisco Mojica: Discovering CRISPR in bacteria

Francisco Mojica.

Similar to other DNA-cutting enzymes involved in DNA repair (described in Concept 16.2), Cas9 is a nuclease that cuts double-stranded DNA molecules. The power of Cas9 for gene editing is that the Cas9 protein will cut any sequence to which it is directed. Cas9 is directed to its target by a guide RNA molecule that it binds and uses as a homing device, cutting both strands of any DNA sequence that is complementary to the guide RNA. Scientists have been able to exploit the function of Cas9 by introducing a Cas9–guide RNA complex into a cell they wish to alter **(Figure 17.28)**. The guide RNA in the complex is engineered to be complementary to the "target" gene. Cas9 cuts both strands of the target DNA, and the resulting broken ends of DNA trigger a DNA repair system (similar to that shown in Figure 16.20). When there is no undamaged DNA for the enzymes of the repair system to use as a template, as shown at the bottom left of Figure 17.28, the repair enzymes introduce or remove random nucleotides while rejoining the ends. Generally, this process alters the DNA sequence so that the gene no longer works properly. This technique is a highly successful way for researchers to "knock out" (disable) a given gene to study what that gene does in an organism.

Figure 17.28

Gene editing using the CRISPR-Cas9 system.

Gene editing using the C R I S P R-C a s 9 system.

Guide R N A engineered to guide the Cas9 protein to a target gene. It has a complementary sequence that can bind to a target gene. The Cas9 protein has active sites that can cut D N A.

1. C a s 9 protein and guide R N A are allowed to bind to each other, forming a complex that is then introduced into a cell.
2. In the nucleus, the complementary sequence of the guide R N A binds to part of the target gene. The active sites of the C a s 9 protein cut the D N A on both strands.
3. The broken strands of D N A are repaired by the cell in one of two ways. First. Scientists can disable (knock out) that target gene to study its normal function. No template is provided, and repair enzymes insert and, or delete random nucleotides, making the gene nonfunctional. Or second. If the target gene has a mutation, it can be repaired by providing a normal copy of the gene. Repair enzymes use the normal gene as a template and synthesize the correct gene sequence.

But what about using this system to help treat genetic diseases? Researchers have modified the technique so that the CRISPR-Cas9 system can be used to repair a gene that has a harmful mutation. They introduce a segment from the normal (functional) gene along with the CRISPR-Cas9 system. After Cas9 cuts the target DNA, repair enzymes can use the normal DNA as a template to repair the target DNA at the break point. In this way, the CRISPR-Cas9 system edits the defective gene so that it is corrected (see the bottom right of Figure 17.28).

In 2018, researchers reported promising results using the CRISPR-Cas9 system in an attempt to correct the genetic defect that causes sickle-cell disease. They edited human cells from patients with sickle-cell disease and injected them into bone marrow in mice. After 19 weeks, the gene remained corrected in 20-40% of the injected cells. A 2021 study using an improved CRISPR-based technique in a similar experiment showed the gene to be altered in roughly 70% of cells, 16 weeks later. Researchers, physicians, and patients alike are excited about the potential of CRISPR technology to treat or even cure human diseases that have a genetic basis, such as sickle-cell, Alzheimer's, and Parkinson's diseases, as well as some types of cancer. However, there are still concerns about using CRISPR-treated cells in humans because of possible effects on genes that are not being targeted; this is an active area of research. The risks associated with cutting both strands of DNA by CRISPR are avoided when CRISPR is used in conjunction with a new technique called *base editing*. In this process, specific bases in a gene are not cut and replaced, but rather chemically modified to resemble a different base, thus altering the resulting protein so it is no longer harmful.

Jennifer Doudna, a co-discoverer of CRISPR-Cas9, recognized not only its incredible potential but also the danger of its misapplication. After having a dream in which Adolf Hitler asked her what could be done with this gene-editing tool, Doudna realized it was imperative to spend some time reflecting on ethical considerations. In 2015, she convened a conference of biologists who agreed at the end of the meeting to urge extreme caution as the field moves forward—and the discussion continues.

What Is a Gene? *Revisiting the Question*

Our definition of a gene has evolved over the past few chapters, as it has through the history of genetics. We began with the Mendelian concept of a gene as a discrete unit of inheritance that affects a phenotypic character (Chapter 14). We saw that Morgan and his colleagues assigned such genes to specific loci on chromosomes (Chapter 15). We went on to view a gene as a region of specific nucleotide sequence along the length of the DNA molecule of a chromosome (Chapter 16). Finally, in this chapter, we have considered a functional definition of a gene as a DNA sequence that codes for a specific polypeptide chain or a functional RNA molecule, such as a tRNA. All these definitions are useful, depending on the context in which genes are being studied.

We have noted that merely saying a gene codes for a polypeptide is an oversimplification. Most eukaryotic genes contain noncoding segments (such as introns), so large portions of these genes have no corresponding segments in polypeptides. Molecular biologists also often include promoters and certain other regulatory regions of DNA within the boundaries of a gene. These DNA sequences are not transcribed, but they can be considered part of the functional gene because they must be present for transcription to occur. Our definition of a gene must also be broad enough to include the DNA that is transcribed into rRNA, tRNA, and other RNAs that are not translated. These genes have no polypeptide products but play crucial roles in the cell. Thus, we arrive at the following definition: *A gene is a region of DNA that can be expressed to produce a final functional product that is either a polypeptide or an RNA molecule*.

When considering phenotypes, however, it is often useful to start by focusing on genes that code for polypeptides. In this chapter, you have learned in molecular terms how a typical gene is expressed—by transcription into RNA and then translation into a polypeptide that forms a protein of specific structure and function. Proteins, in turn, bring about an organism's observable phenotype.

A given type of cell expresses only a subset of its genes. This is an essential feature in multicellular organisms: You'd be in trouble if the lens cells in your eyes started expressing the genes for hair proteins, which are normally expressed only in hair follicle cells! Gene expression is precisely regulated, which we'll explore in the next chapter, beginning with the simpler case of bacteria and continuing with eukaryotes.